FLEET TEAM DIGEST

737MAX-FTD-32-17015

Issue Title : 737NG Experience With Low/Flat Nose Landing Gear Shock Strut

Airplane Model	ATA	Minor Model(s)
737MAX	3220-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
05/05/2017	05/05/2017	04/18/2017

Estimated Completion Date	Next Update Date

Applicability

All 737MAX models

Description

This FTD article discusses fluid and/or nitrogen leak occurrences and valve replacements that may occur on the 737 MAX.

A number of 737NG operators have reported that the nose landing gear shock strut is repeatedly being seen as either low or flat, but that there is no evidence of a fluid leak.

There are a number of places where fluid and/or nitrogen can escape from the sealed shock strut. These are:

Air Charge Valve (Schrader Valve P/N MS28889-2)

Fluid Charge Valve (Check Valve P/N 2C9342)

Dynamic Seal

Static Seal

Metering Pin Seal

Orifice Support Tube Seal

The attachment to this FTD titled 'Nose Landing Gear Leak Paths' pictorially shows where fluid and/or nitrogen can escape from the shock strut.

For fluid leaks, the source of the leak can be easily determined but for a nitrogen leak, it can be very difficult to find the source.

We have experienced a 737NG shock strut that kept going low/flat, without evidence of a fluid leak. Trouble shooting steps included removing the cap (P/N BACC14AD04N) from the check valve (P/N 2C9342) while the strut was still pressurized. When this was done, a small amount of fluid/nitrogen mix (emulsion) slowly escaped from the valve for a very short period of time. See the file 'NLG Check Valve Leaking' attached to this FTD article. It is thought that with the cap installed, this intermittent leak

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caused nitrogen to escape through the cap but that the cap kept fluid from escaping. From all the troubleshooting conducted, there were no other findings that would indicate the source of a leak. After replacing all of the seals in the shock strut, and installing a new check valve, the strut performed normally.

Status

Boeing will monitor the fleet data for reports of low and depressurized shock struts.

Interim Action

None

Final Action

None

Operator Action

We recommend that when a nose gear shock strut is reported as low, the pressure and x-dimension be checked to verify that the strut is indeed 'off the chart' prior to making any adjustments. Note that due to the very small volume nose gear strut, with a forward CG and relatively heavy aircraft, the nose gear strut will look low, but it may indeed still be on the servicing curve. Service as necessary with nitrogen in accordance with AMM 12-15-41. If a knocking sound is reported, then the strut is likely low and should be checked for leaks and correct servicing.

To verify that the correct fluid volume is in the strut, a 2 point pressure check should be performed at 2 different aircraft weights (or with the aircraft jacked, 2 different x-dimensions can be achieved by moving the inner cylinder with an axle jack). The strut must be on the servicing band at both weights/x-dimensions. It is acceptable to simply top off with fluid in lieu of performing a 2 point check to ensure that the correct fluid volume is in the strut.

Note: AMM 12-15-41 requires a 2 inch difference in x-dimension to perform the 2 point pressure check, which is difficult to achieve on the nose gear within the normal operating weight envelope. As long as 2 different x-dimensions can be achieved, the check will indicate if the correct fluid volume is in the strut.

If the strut continues to be reported low (or flat) and there is no evidence of a fluid leak from any valve or seal (especially the dynamic seal), we recommend replacing the air charging valve P/N MS28889-2 (Schrader Valve) and/or the fluid charge valve P/N 2C9342 (Check Valve) to eliminate those items as possible sources of nitrogen leaks. Both valves can be checked with Snoop prior to replacement, but in our experience with the leaking check valve, the leak can apparently be intermittent.

We also believe, for operators experiencing frequent reports of low struts, that it could be beneficial to perform a full fluid and nitrogen service more frequently than the MPD Task 32-070-00 recommended 3000 cycle interval. After performing a full fluid and nitrogen service, the strut pressure and x-dimension should be checked within 5-10 cycles in accordance with AMM 12-15-41.

If an operators' reliability data indicates that low strut reports are seasonal, and they are more frequent during winter months, it is acceptable to over-service the strut with additional nitrogen to increase the x-dimension by 1 inch. This should be done in a relatively warm environment (e.g. warm weather or after resting in an ambient temperature hangar) such that when the aircraft is cold, or arrives in a cold environment, the strut is at a more normal x-dimension. This procedure is described in the Extreme Cold Weather Maintenance AMM section, 12-33-02.

Part Information

Check Valve P/N 2C9342

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Schrader Valve P/N M28889-2

NLG Component Installation P/N 162A1100-x

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Aircraft Maintenance Manual (AMM)	12-15-41		
Aircraft Maintenance Manual (AMM)	12-33-02		
Maintenance Planning Document (MPD)	32-070-00		

Related Categories

- Airplane Maintenance Issues : Issues that occur during airplane maintenance or have an impact on airplane reliability
- Dispatch Reliability Issues : Issues that cause revenue flight delays and cancellations
- In Service Occurrences : null

Attachments:

a) NLG Check Valve Leaking

b) Nose Landing Gear Leak Paths

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